

Text-to-Metaverse: Towards a Digital Twin-Enabled Multimodal Conditional Generative Metaverse

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ABSTRACT

Developing realistic and interactive virtual environments is a major hurdle in the progress of Metaverse. At present, majority of Metaverse applications necessitate the manual construction of 3D models which is both time-consuming and costly. Additionally, it is challenging to design environments that can promptly react to users' actions. To address this challenge, this paper proposes a novel approach to generate virtual worlds using digital twin (DT) technology and AI through a Text-to-Metaverse pipeline. This pipeline converts natural language input into a scene JSON, which is used to generate a 3D virtual world using two engines: Generative Script Engine (GSE) and Generative Metaverse Engine (GME). GME generates a design script from the JSON file, and then uses it to generate 3D objects in an environment. It aims to use multimodal AI and DT technology to produce realistic and highly detailed virtual environments. The proposed pipeline has potential applications including education, training, architecture, healthcare and entertainment, and could change the way designers and developers create virtual worlds. While this short paper covers an abstract as per the Doctorial Symposium's guidelines, it contributes to the research on generative models for multimodal data and provides a new direction for creating immersive virtual experiences.

CCS CONCEPTS

• **Computing methodologies** → *Computer vision; Natural language processing.*

KEYWORDS

Metaverse, Digital Twin, Multimodal AI, Computer Vision, NLP, Generative Models.

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1 INTRODUCTION

The concept of metaverse, a virtual universe that simulates a physical world or a fictional one, has gained significant attention in recent years due to the growth of virtual reality (VR) and augmented reality (AR) technologies [12]. Metaverse provides a shared space for users to interact with each other and the environment, creating new opportunities for socialization, entertainment, and commerce [9]. One of the challenges in the development of the Metaverse is the creation of realistic and interactive virtual environments. Currently, most Metaverse applications rely on manual creation of 3D models, which can be time-consuming and expensive. Moreover, it is difficult to create environments that respond to users' actions in real-time [8]. To address this challenge, we propose a novel approach for creating immersive virtual environments, using text as input. Our approach, Text-to-Metaverse, leverages digital twin (DT) technology [6] and conditional generative adversarial networks (GANs) [4] to generate realistic and interactive Metaverse environments based on textual descriptions.

One promising approach is to use DT technology to generate virtual environments that reflect the properties of the input text. DTs are virtual replicas of physical systems or processes that can simulate their behavior and interactions with the environment [6]. They have been used in various domains, such as manufacturing, healthcare, and energy management, to optimize processes and reduce costs [14]. In the context of virtual environments, DTs can be used to create a high-level representation of the input text and guide the generation of realistic content, such as 3D models, textures, and sounds. GANs are a popular class of AI models that can generate realistic content, such as images, videos, and speech, by learning from a dataset of examples [4]. Recent works have explored the use of GANs for multimodal data, such as images and text [19], as well as speech and text [15]. In the context of virtual environments, GANs can be used to generate realistic visual and auditory content that corresponds to the DT representation of the input text.

This paper proposes a novel method for generating virtual worlds via a Text-to-Metaverse pipeline that employs DT technology and multimodal AI. The pipeline converts natural language input into a JSON file specifying the environment, objects, and their inter-relationships. To create a 3D virtual world, the pipeline employs two engines: the Generative Script Engine (GSE) and the Generative Metaverse Engine (GME). GME uses the JSON file to generate a design script that defines the 3D objects, 3D scene, and their attributes. This script is then used to position and render the objects and their surroundings. By utilizing multimodal AI-based DT, the system can accurately interpret complex natural language descriptions, resulting in immersive virtual environments. The initial

results suggest that this text-to-metaverse pipeline has the potential to transform virtual world creation in many industries including education, training, healthcare, architecture and entertainment.

The rest of the paper is organized as follows: Section 2 reviews the related works on DTs, multimodal GANs and their variants. Section 3 describes our methodology, including the overall Text-to-Metaverse pipeline and its stages. Section 4 provides a walk through the pipeline and illustrates the mechanism with an example. Finally, Section 5 concludes the paper and highlights the future work.

2 RELATED WORK

2.1 Digital Twins and their Applications

A DT is a virtual replica of a physical system or process that allows for simulations, predictions, and analysis of real-world scenarios. DTs have been widely researched and implemented in recent years in various fields, where AI has played a crucial role in enhancing the capabilities of DTs by enabling the development of intelligent algorithms that can learn from data and make predictions [3]. Recent studies have explored the use of DTs in various fields such as healthcare, where they have been used to simulate and predict the behavior of organs and tissues, enabling personalized medicine and treatment planning [1]. In urban planning, DTs have been used to simulate the behavior of traffic, predict energy consumption, and optimize waste management [2].

The use of AI and multimodal approaches in DTs has also enabled the development of novel techniques such as Neural Radiance Fields (NeRF) for scene reconstruction from images [7]. NeRF is a technique that uses neural networks to model the radiance field of a scene, enabling the generation of high-quality 3D models from 2D images. Other techniques such as text-to-image synthesis and video synthesis have also been explored using AI and multimodal approaches [11, 18]. These techniques contributed to enabling the generation of realistic images and videos from textual descriptions, bringing us closer to the metaverse.

2.2 Multimodal GANs and their Variants

Multimodal GANs have emerged as a powerful tool for generating high-quality samples across modalities such as text, image, and video. These models leverage multiple input modalities to generate samples that capture the underlying distribution of the data. Conditional GAN is a popular variant used to generate images from specific input conditions such as text-to-image synthesis [19]. It has been extended to handle multimodal data and have been applied in generating videos from textual descriptions [20], neural radiance fields [7], translating natural language sentences into 3D objects [16], and high-resolution image synthesis using StyleGAN [5].

In recent years, several variants of GANs have been proposed to improve the performance and stability of multimodal GANs. For instance, BicycleGAN [21] uses an encoder to map images to a shared latent space with the textual descriptions, enabling more fine-grained control over the generated images. Another notable variant is the AttnGAN [17], which incorporates attention mechanisms to focus on the relevant parts of the textual descriptions when generating the images. Similarly, MirrorGAN [10] uses a dual attention mechanism to encourage the generator to focus on both the textual descriptions and the real images while generating new samples. Text2Scene [13] was introduced recently as an approach

to generate 2D scenes from text using a neural network. These models have become increasingly powerful and versatile, bringing us one step closer to realizing the potential of the metaverse.

3 METHOD

As per the Doctorial Symposium's paper call, this work is in its early stages with an initial proposed pipeline as in Fig. 1. The following outlines the detailed proposed methodology for a text-to-metaverse pipeline that takes text descriptions as input and generates 3D environments. The pipeline comprises of three stages:

- (1) **Text Input Processing Module**, which extracts *Scene JSON file* specifying the *Environment*, *Objects*, and *Relations*,
- (2) **Generative Script Engine (GSE)**, which creates a design script by scripting 3D *Objects*, *3D Scene*, and *Properties*,
- (3) **Generative Metaverse Engine (GME)**, which executes the design script to perform 3D Object Generation followed by 3D Object Positioning, and Scene Rendering to produce the desired 3D environment. Further details can be found below.

3.1 Text Input Processing Module

The Text Input Processing Module is responsible for extracting relevant information from the text description and converting it into a structured format that can be used by subsequent stages of the pipeline. The following are the steps involved in this process:

- (1) **Text Preprocessing**: The text is preprocessed to remove any unwanted characters, stop words, and punctuation marks. The text is then tokenized, and each token is assigned a part of speech tag with NLP.
- (2) **Entity Recognition**: The text is parsed to identify the entities mentioned in it. Entities can be objects, locations, or characters. This is done using a trained NLP model that recognizes named entities.
- (3) **Relation Extraction**: The text is analyzed to identify the relationships between the entities mentioned in it. The relations can be spatial, temporal, or causal.
- (4) **Scene JSON Generation**: Using the information extracted from the previous steps, the Scene JSON is generated including information of the Environment, Objects, and Relations.

3.2 Generative Script Engine (GSE):

The GSE is responsible for creating a design script that can be used to generate a 3D environment based on the Scene JSON generated in the previous step. The following are the steps of this process:

- (1) **Object Scripting**: The 3D objects that need to be created in the scene are scripted using a scripting language like Python. This includes defining the size, shape, texture, and position of the objects.
- (2) **Scene Scripting**: The 3D scene is scripted based on the Scene JSON generated in the previous step. This includes defining the camera position, lighting, and background.
- (3) **Property Scripting**: The properties of the objects and the scene are scripted. This includes defining the color, texture, and other visual attributes of the objects.
- (4) **Design Script Generation**: The Object Script, Scene Script, and Property Script are combined to generate a design script that can be executed by the GME.

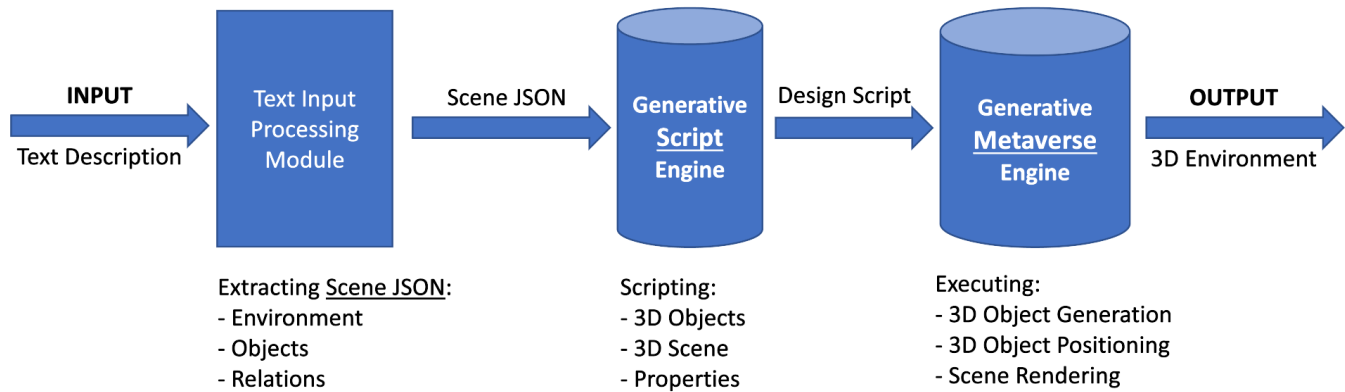


Figure 1: Overview of the Text-to-Metaverse pipeline, depicting the three stages involved in generating a 3D virtual environment from a natural language text input. The input text is processed by the Text Input Processing Module to extract a Scene JSON comprising the Environment, Objects, and Relations. The GSE then scripts the 3D Objects, 3D Scene, and Properties to produce a design script, which is used by the GME to execute 3D Object Generation, 3D Object Positioning, and Scene Rendering to output the final 3D environment.

3.3 Generative Metaverse Engine (GME):

The GME is responsible for executing the design script generated by the GSE to produce the final 3D environment. The following are the steps involved in this process:

- (1) **Object Generation:** The 3D objects are generated based on the Object Script in the design script.
- (2) **Object Positioning:** The 3D objects are positioned in the scene based on their position in the Scene Script.
- (3) **Scene Rendering:** The scene is rendered based on the design script, including the objects and properties defined in the design script.
- (4) **Output Generation:** The final 3D environment is generated as an output of the pipeline, which can be visualized in a virtual reality environment or a rendering software.



Figure 2: Example of the aimed generated rendered output of the pipeline using the text input "generate a 3D scene of a human surrounded by a tree and three balls next to a cat on a vertical square stage in an environment of various colors".

4 WALK THROUGH THE PIPELINE

Providing a walk through the pipeline mechanism, Fig. 2 shows an example of the aimed generated rendered output of the text-to-metaverse pipeline given the text input "generate a 3D scene of a human surrounded by a tree and three balls next to a cat on a vertical square stage in an environment of various colors". First, the Text Input Processing Module extracts the following Scene JSON:

```

1 {"environment": {
2   "color": "various"},
3  "objects": [
4    {"type": "human"},
5    {"type": "tree"},
6    {"type": "ball",
7     "count": 3},
8    {"type": "cat"},
9    {"type": "stage",
10     "shape": "square",
11     "orientation": "vertical"}],
12 "relations": [
13   {"subject": "human",
14    "predicate": "surrounded by",
15    "object": "tree"},
16   {"subject": "human",
17    "predicate": "surrounded by",
18    "object": "ball"},
19   {"subject": "human",
20    "predicate": "next to",
21    "object": "cat"},
22   {"subject": "cat",
23    "predicate": "on",
24    "object": "stage"}]]
  
```

Algorithm 1 Pseudo code of the "Design Script" for the GME based on the generated "Scene JSON" from the GSE. This script can be implemented and rendered with further details using Python in Blender or C# in Unity 3D for instance.

```

1: for object in objects do
2:   if object.type == "human" then
3:     Create human object           ▶ Save reference as human
4:   else if object.type == "tree" then
5:     Create tree object           ▶ Save reference as tree
6:   else if object.type == "ball" then
7:     Create ball object           ▶ Save reference as ball1
8:     Duplicate ball two more times ▶ Save as ball2 and ball3
9:   else if object.type == "cat" then
10:    Create cat object             ▶ Save reference as cat
11:   else if object.type == "stage" then
12:    Create stage object           ▶ Save reference as stage
13:    if object.shape == "square" then
14:      Set stage shape to square
15:    end if
16:    if object.orientation == "vertical" then
17:      Set stage orientation to vertical
18:    end if
19:  end if
20: end for
21: Set environment color to environment.color
22: Position human next to cat
23: Position cat on stage
24: Position human surrounded by tree and ball1, ball2, ball3

```

5 CONCLUSION AND PLANNED WORK

In conclusion, this paper presents to ACM MM's Doctorial Symposium a novel direction for generating virtual worlds via a Text-to-Metaverse pipeline that employs DT technology and multimodal AI. The proposed pipeline has the potential to revolutionize virtual world creation and exploration in various industries such as education, training, architecture, gaming, and film. However, there are limitations to this work, such as the need for further evaluation and improvement of the system's performance in generating more complex and realistic virtual worlds. Planned future work will focus on automating the pipeline to be end-to-end for both training and inference, incorporating a wide database of DT objects to serve each field sufficiently, integrating state-of-the-art deep learning generative models, expanding the capabilities of the pipeline to support user customization and interaction with virtual worlds, creating more robust and versatile design engines, and exploring the potential ethical implications of using such technology. Overall, the proposed pipeline presents a promising avenue for advancing the field of virtual world creation and exploring new possibilities for immersive experiences.

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